

Mahia Akter Momo

Machine Learning Engineer

✉ mahiamomo12@gmail.com ☎ +8801850127053 📍 Uttara Dhaka, Bangladesh

🔗 mahiamomo-portfolio.vercel.app 🔗 linkedin.com/in/mahiamomo12/ 🔗 github.com/mahiamOmO

👤 PROFILE

Computer Science and Engineering graduate with hands-on experience in **Artificial Intelligence, Machine Learning, Computer Vision**, and **AI Data Annotation**. Experienced in **data annotation, dataset preparation, quality assurance, model evaluation**, and data-centric improvement for AI/ML projects. Skilled in **Python, PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, OpenCV**, and **Hugging Face**. Interested in building and improving reliable machine learning systems through high-quality data and practical AI solutions.

📁 PROFESSIONAL EXPERIENCE

- ELITE Research Lab LLC, Student Researcher**1 Sept 2026 – present
Queens, New York, USA
- Conducting research in Computer Vision, NLP, Deep Learning, LLMs, and Generative AI.
 - Performing literature reviews and identifying research gaps and potential research directions.
 - Working on dataset preparation, model development, training, fine-tuning, and evaluation.
 - Conducting experiments and analyzing results to support research projects and publications.
- Agency Handy, (Intern) AI Data Annotation & Quality**Dec 1 – Mar 31 2026
Uttara, Dhaka, Bangladesh
- Supporting AI and Machine Learning projects by accurately annotating and reviewing data.
 - Ensuring high-quality labeled datasets for model training and improvement.
 - Maintaining accuracy, consistency, and quality across annotated datasets according to project guidelines.
 - Preparing and validating annotated datasets to support reliable AI/ML model development and evaluation.

🏆 COMPETITION PARTICIPATION

- Solvio AI Hackathon, Sheba Platform Ltd**11/2025
(3,259+ teams, securing a top 394 rank)
- Technovation25 Hackathon, Josephite IT Club**09/2025
(Top 10 in the Final Round)
- INNOVATEX 2025 Hackathon, University of Asia Pacific**05/2025
(Top 20 in the Final round)
- IDEA Season 4.0, Entrepreneurship and Career Development Club of Uap -ECDC**04/2025
(IDEA Season 4.0)
- UAP Intra University Collaborative Programming Contest 1.0, University of Asia Pacific**01/2024
(Top 40 out of 70+ teams)

SKILLS

Programming Language

Python, MySQL, NumPy, Pandas, Data Cleaning, Data Preprocessing, JSON/CSV

Backend & Deployment

FastAPI, Flask, REST API, Jupyter Notebook

Generative AI & LLM

Hugging Face, OpenAI API, LangChain, RAG, Prompt Engineering

Annotation Tools

SuperAnnotate, Supervisely, Superb AI, LabelImg, Label Studio, CVAT

Computer Vision & AI

Computer Vision, OpenCV, Object Detection, Image Classification, Image Processing

AI Data Annotation

Data Annotation, Image & Video Annotation, Bounding Box & Polygon Annotation, Dataset Preparation, Data Validation, QA

Machine Learning

Scikit-learn, XGBoost, PyTorch, TensorFlow, Machine Learning, Deep Learning, Model Evaluation

PROJECTS

Failure-Risk-Prediction , *TPS ML Web App*

Failure Risk Prediction is a web application that predicts product failure risk using the TPS dataset and an XGBoost model. Built with Flask, it provides a user-friendly interface to input product parameters and get real-time risk predictions.

Backend: Python, Flask, Flask-CORS

Machine Learning: XGBoost, Scikit-learn, Joblib

Data Processing: Pandas, NumPy

Frontend: HTML, JavaScript, CSS

Daily-task-prioritization-agent , *AI Agent*

Daily Task Prioritization Agent is an intelligent full-stack task management application that uses AI to prioritize and manage daily tasks. It features a modern Next.js frontend and a scalable FastAPI backend, offering smooth task handling and responsive design.

Frontend:

Next.js, TypeScript, Tailwind
CSS, shadcn/ui, Lucide Icons

Backend:

FastAPI, Python, Uvicorn,
Pydantic, CORS Middleware

Books Vibes , *FullStack*

Book Vibes is a Django-based online bookstore where users can browse books, add them to their cart, and place orders seamlessly.

Backend: Python, Django, SQLite **Frontend:** HTML, CSS, Javascript, Bootstrap

UAP-Alumni-Connect, *FullStack*

UAP Alumni Connect is an online platform connecting UAP alumni, students, and faculty through profile browsing, networking, and community engagement. **Frontend:** React.js, HTML, CSS, JavaScript, Firebase Auth | **Backend:** Supabase, PostgreSQL, Node.js, Express.js

EDUCATION

B.Sc in Computer Science and Engineering, *University of Asia Pacific*

07/2022 – 07/2026